PROJECT PROPOSAL

Fasal Sehat – Smart Agricultural Assistance System

Course: Software Construction and Development

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1. Project Overview

Fasal Sehat is a smart agricultural assistance system designed to help farmers make better decisions about crop health, farming practices, and common agricultural problems. The system provides organized guidance and aims to make useful agricultural information easier to access.

2. Problem Statement

Farmers may face difficulty identifying crop-related problems and finding reliable, timely guidance. Traditional methods can depend heavily on personal experience or manual consultation. Fasal Sehat proposes a software-based solution that organizes agricultural information and assists users in making informed decisions.

3. Objectives

- Provide an easy-to-use platform for agricultural assistance.
- Help users identify common crop and farming problems.
- Provide relevant recommendations and guidance.
- Organize agricultural information in a structured and reusable way.
- Apply software construction principles such as modularity, encapsulation, abstraction, and reusability.
- Create a maintainable system that can be extended with additional crops, diseases, and recommendations.

4. Proposed Features

- User-friendly interface for farmers.
- Crop selection and crop-related information.
- Identification of common crop problems.
- Agricultural recommendations and preventive guidance.
- Search and retrieval of relevant information.
- Structured storage of crop, disease, and recommendation data.
- Separate modules for user interaction, processing, and data management.
- Validation of user input and clear error messages.

5. Software Construction Approach

The project will be developed using a modular object-oriented approach. Responsibilities will be separated into appropriate classes and components instead of placing all logic in one main program. Private fields, methods, constructors, validation, and reusable service classes will be used where appropriate.

- Modularity – divide the application into focused components.
- Encapsulation – protect internal data and expose controlled operations.
- Abstraction – hide unnecessary implementation details from users and calling components.
- Reusability – design services and utility methods that can be reused across the application.
- Maintainability – keep responsibilities clear so future changes can be made with less risk.
- Testing – test normal, boundary, and invalid inputs.

6. Major System Components

- User Interface Module – handles user interaction and input.
- Crop Information Module – manages crop-related information.
- Problem/Disease Module – stores and retrieves information about common agricultural problems.
- Recommendation Module – provides suitable guidance based on the selected problem or crop.
- Data Management Module – manages structured project data.
- Validation Module – checks user input and prevents invalid data from entering the system.

7. Expected Benefits

- Makes agricultural information easier to access.
- Reduces dependence on scattered or unorganized information.
- Provides structured guidance for common farming problems.
- Demonstrates practical application of software construction concepts.
- Provides a foundation that can be expanded in future versions.

8. Testing Plan

Testing will be performed at component and system level. The project will include normal test cases, boundary cases, and invalid-input cases. Each major module will be tested independently before integration, followed by end-to-end testing of the complete system.

9. Development Tools

- Programming Language: Java
- Development Environment: NetBeans / IntelliJ IDEA / Eclipse
- Version Control: Git (if required)
- Database or structured data storage: selected according to project implementation requirements

10. Conclusion

Fasal Sehat is proposed as a practical software construction project that combines a real-world agricultural problem with sound software engineering practices. The project will focus on modular design, object-oriented programming, validation, testing, and maintainability while providing useful agricultural assistance to its intended users.

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